Licenciatura em Engenharia Informática
ALGAN
1º Semestre 2025-2026
TP1

1) Write the matrices defined by::

(a) $\mathbf{A} = [a_{ij}]_{\substack{1\leq i\leq 3\\ 1\leq j\leq 2}}$ where $a_{ij} = 2i - 3j$

(b) $\mathbf{B} = [b_{ij}]_{\substack{1\leq i\leq 4\\ 1\leq j\leq 4}}$ where $b_{ij} = \frac{1}{i+j}$

(c) $\mathbf{C} = [c_{ij}]_{\substack{1\leq i\leq 2\\ 1\leq j\leq 3}}$ where $c_{ij} = 5 - i^j$

(d) $\mathbf{D} = [d_{ij}]_{\substack{1\leq i\leq 3\\ 1\leq j\leq 3}}$ where $d_{ij} = \sqrt{ij}$

(e) $\mathbf{E} = [e_{ij}]_{\substack{1\leq i\leq 4\\ j=1}}$ where $e_{ij} = (-2)^i(j-3)$

(f) $\mathbf{F} = [f_{ij}]_{\substack{1\leq i\leq 2\\ 1\leq j\leq 3}}$ where $f_{ij} = \begin{cases} 1 & \text{se } i = j \\ i - j & \text{se } i \neq j \end{cases}$

2) Considering the matrices from the previous exercise perform, if possible, the following matrix operations. Justify the impossibility.

(a) $\mathbf{A} + \mathbf{C}$

(b) $\mathbf{C} - 2\mathbf{F}$

(c) $\mathbf{A}^T\mathbf{D}$

(d) $\mathbf{CA}$

(e) $\mathbf{CD} + \mathbf{A}$

3) Let $\mathbf{A}$ be a matrix of order 3 where $a_{ij} = i^2 - j$. The product of the elements of the secondary diagonal of matrix $\mathbf{A}$ is:

a) 0 b) 16 c) -32

d) 32

4) If $\mathbf{B} = [b_{ij}]_{\substack{1\leq i\leq 4\\ 1\leq j\leq 4}}$ where $b_{ij} = \begin{cases} 0 & \text{if } i = j \\ i - j & \text{if } i \neq j \end{cases}$ then:

a) $\mathbf{B}$ is symmetric b) $\mathbf{B} + \mathbf{B}^T = \mathbf{I}_4$ c) $\mathbf{B}$ is skew-symmetric

d) $\mathbf{B} - \mathbf{B}^T = 0_4$

5) the matrix $\mathbf{C} = \begin{bmatrix} 4+a & a+b \\ 2(a-1)-b & b \end{bmatrix}$ is a diagonal matrix if:

a) $a = b = \frac{2}{3}$ b) $a = b = -\frac{2}{3}$ c) $a = \frac{2}{3}$ e $b = -\frac{2}{3}$

d) $a = -\frac{2}{3}$ e $b = \frac{2}{3}$

jem@isep.ipp.pt

6) Consider the matrices

$$\mathbf{A} = \begin{bmatrix} 1 & -2 & -1 & 0 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & -2 & -1 & 0 \\ 0 & -2 & 1 & 0 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 0 & -2 & 2 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 \end{bmatrix}$$

$$\mathbf{D} = \begin{bmatrix} 0 \\ -1 \\ 3 \\ 0 \end{bmatrix} \qquad \mathbf{E} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} \qquad \mathbf{F} = \begin{bmatrix} 0 & 1 & 2 & 1 \\ 0 & -2 & -1 & 0 \\ 2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Compute, if possible, the following matrices:

a) $\mathbf{A}\,\mathbf{B}$	**b)** $\mathbf{B}\,\mathbf{A}$	**c)** $\mathbf{A}\,\mathbf{B}^{\mathbf{T}}$	**d)** $\mathbf{B}\,\mathbf{B}^{\mathbf{T}}$	**e)** $\mathbf{B}\,\mathbf{I_2}$
f) $\mathbf{C} + 2\mathbf{D}^{\mathbf{T}}$	**g)** $\mathbf{B}\,(\mathbf{C} + \mathbf{F})$	**h)** $(\mathbf{C} + \mathbf{F})\,\mathbf{B}$	**i)** $(\mathbf{C} + \mathbf{F})\,\mathbf{B}^{\mathbf{T}}$	**j)** $\mathbf{I_2}\,(\mathbf{B} - 3\mathbf{E})$
k) $4\mathbf{A}^{\mathbf{T}} + \mathbf{D}$	**l)** $\mathbf{C}\,\mathbf{A}^{\mathbf{T}}$	**m)** $(\mathbf{C}\,\mathbf{A})^T$	**n)** $(\mathbf{C} - \mathbf{F})^2$	**o)** $(\mathbf{B}\mathbf{E})^2$
p) $(\mathbf{E}\mathbf{B})^2$	**q)** $\mathbf{B}\,\mathbf{C}\,\mathbf{F}\,\mathbf{D}$	**r)** $\mathbf{A}\,\mathbf{A}^{\mathbf{T}}$	**s)** $\mathbf{E}\,\mathbf{B} + \mathbf{C}$	**t)** $(\mathbf{C}\,(\mathbf{D} + \mathbf{A}^{\mathbf{T}}))^T\,\mathbf{E}$

7) Consider the matrices $\mathbf{A}$ e $\mathbf{B} \in \mathcal{M}_n$. Under what conditions is the relation verified?,

$$(\mathbf{A} + \mathbf{B})^2 = \mathbf{A}^2 + 2\,\mathbf{A}\,\mathbf{B} + \mathbf{B}^2$$

8) Let $\mathbf{A}$ e $\mathbf{B} \in \mathcal{M}_n$ symmetric matrices. Under what conditions the matrix $\mathbf{A}\,\mathbf{B}$ is symmetric?

9) Solve the following matricial equations for the unknown $\mathbf{X}$.

a) $\mathbf{A}\,\mathbf{X} + \mathbf{X}\,\mathbf{B} = \mathbf{C}$, where $\mathbf{A} + \mathbf{B}$ is invertible and $\mathbf{X}$ commute with $\mathbf{A}$.

b) $\mathbf{A}\,\mathbf{X} + \mathbf{X}\,\mathbf{B} = \mathbf{C}$, where $\mathbf{A} + \mathbf{B}$ invertiblel and $\mathbf{X}$ commute with $\mathbf{B}$.

c) $(\mathbf{A}^T\,\mathbf{X})^T - \mathbf{I}^5 = \mathbf{0}$, where $\mathbf{A}$ is invertible.

d) $((\mathbf{A}^T\mathbf{X})\mathbf{B})^T = (\mathbf{A}^T)^{-1}$, where $\mathbf{A}$, $\mathbf{B}$ are invertible and $\mathbf{A}^{-1}$ is symmetric.

e) $(\mathbf{B}^T\mathbf{X}^{-1})^{-1}\mathbf{B}^T\mathbf{C} + (\mathbf{C}^T\mathbf{B}^{-1})^T = \mathbf{I}$, where $\mathbf{C}$ is invertible.

f) $\mathbf{A}(\mathbf{B} + \mathbf{X})\mathbf{B}^{-1} = \mathbf{I}$ where the matrices involved are invertible.

g) $[\mathbf{A}(\mathbf{B} + \mathbf{X})]^T = \mathbf{I}$, where the matrices involved are invertible.

10) Find the set of matrices that commute with the matrix $\begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}$.

11) Consider the matrix $\mathbf{B} = \begin{bmatrix} -1 & -1 \\ 0 & 2 \end{bmatrix}$.

a) Find the matrix $\mathbf{A}$ that satisfy $\mathbf{A}\,\mathbf{B} = \mathbf{I_2}$.

b) If $\mathbf{B}^{20} = \begin{bmatrix} 1 & -349525 \\ 0 & 1048576 \end{bmatrix}$, find the matrix $\mathbf{B}^{19}$.

12) consider the matrix $\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$.

a) Show that $\mathbf{A}^{\mathbf{n}} = \begin{bmatrix} 1 & -n \\ 0 & 1 \end{bmatrix}$, $\forall n \in \mathbb{N}$.

b) Find it inverse, $\mathbf{A^{-1}}$.

c) Show that $\mathbf{A^n} = \begin{bmatrix} 1 & -n \\ 0 & 1 \end{bmatrix}$, $\forall n \in \mathbb{Z}$.

13) Consider the following matrices:

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 0 & 0 \\ 3 & 1 & 2 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\mathbf{D} = \begin{bmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 2 \end{bmatrix} \quad \mathbf{E} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \end{bmatrix} \quad \mathbf{F} = \begin{bmatrix} 1 & 1 & 24 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 12 & 1 \\ 0 & 0 & 24 & 2 \end{bmatrix}$$

For each matrix mentioned above,

a) Find an equivalent row echelon form, $E(*)$.

b) Find the rank.

14) Find, if possible, two invertible matrices $\mathbf{A}$, $\mathbf{B}$ ($\mathbf{A} \neq -\mathbf{B}$) such that $\mathbf{A} + \mathbf{B}$ is not invertible.

15) Let $\alpha \in \mathbb{R} \setminus \{0\}$ and $\mathbf{A} \in \mathcal{M}_n$ invertible.

a) Show that $\alpha\,\mathbf{A}$ is invertible.

b) Determine $(\alpha\,\mathbf{A})^{-1}$.

16) Consider the matrix $\mathbf{A} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$. Find a matrix $\mathbf{X}$ that satisfy the equation

$$\left[\mathbf{A}^T\mathbf{X}^{-1}\right]^T = (\mathbf{A}^T)^{-1}.$$

17) Consider the matrix $\mathbf{A} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$.

a) Compute $\mathbf{A}^3$.

b) Verify, without computing the inverse matrix, whether $(\mathbf{A}^2)^{-1} = (\mathbf{A}^T)^2$.

c) Verify, without computing the inverse matrix, whether $(\mathbf{A}^n)^{-1} = (\mathbf{A}^T)^n$ for every positive integer n. Hint: Use the identities $(\mathbf{A}^T)^n = (\mathbf{A}^n)^T$ e $\mathbf{A}^n(\mathbf{A^T})^n = \mathbf{A}^{n-1}\mathbf{A}\mathbf{A^T}(\mathbf{A^T})^{n-1}$.

18) Consider the matrices

$$\mathbf{A} = \begin{bmatrix} -2 & -2 & 0 \\ -1 & 1 & 2 \\ 1 & -1 & 2 \end{bmatrix} \text{ e } \mathbf{J} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

a) Use the Gauss–Jordan algorithm to compute $\mathbf{A}^{-1}$.

b) Determine, if possible, a matrix $\mathbf{B}$ such that $\mathbf{A}\mathbf{B}\mathbf{A}^{-1} = \mathbf{J}$.

Solutions

1.

(a) $\mathbf{A} = \begin{bmatrix} -1 & -4 \\ 1 & -2 \\ 3 & 0 \end{bmatrix}$ (b) $\mathbf{B} = \begin{bmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{5} & \frac{1}{6} & \frac{1}{7} \\ \frac{1}{5} & \frac{1}{6} & \frac{1}{7} & \frac{1}{8} \end{bmatrix}$ (c) $C = \begin{bmatrix} 4 & 4 & 4 \\ 3 & 1 & -3 \end{bmatrix}$

(d) $\mathbf{D} = \begin{bmatrix} 1 & \sqrt{2} & \sqrt{3} \\ \sqrt{2} & 2 & \sqrt{6} \\ \sqrt{3} & \sqrt{6} & 3 \end{bmatrix}$ (e) $\mathbf{E} = \begin{bmatrix} 4 \\ -8 \\ 16 \\ -32 \end{bmatrix}$ (f) $\mathbf{F} = \begin{bmatrix} 1 & -1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$

2.

(a) Not possible

(b) $\begin{bmatrix} 2 & 6 & 8 \\ 1 & -1 & -1 \end{bmatrix}$

(c) $\begin{bmatrix} -1+\sqrt{2}+3\sqrt{3} & 2-\sqrt{2}+3\sqrt{6} & 9-\sqrt{3}+\sqrt{6} \\ -4-2\sqrt{2} & -4(\sqrt{2}+1) & -4\sqrt{3}-2\sqrt{6} \end{bmatrix}$

(d) $\begin{bmatrix} 12 & -24 \\ -11 & -14 \end{bmatrix}$

(e) Not possible

3. c) 4.c) 5.c)

6.

It is not possible to perform the operations indicated in : **a)**, **b)**, **e)**, **f)**, **h)**, **j)** and **m)**.

c) $\begin{bmatrix} 6 & 3 \end{bmatrix}$ **d)** $\begin{bmatrix} 6 & 3 \\ 3 & 5 \end{bmatrix}$ **g)** $\begin{bmatrix} -4 & 7 & 4 & 1 \\ 0 & 8 & 0 & 0 \end{bmatrix}$

i) $\begin{bmatrix} -2 & 6 \\ 9 & 8 \\ 2 & 0 \\ -4 & -1 \end{bmatrix}$ **k)** $\begin{bmatrix} 4 \\ -9 \\ -1 \\ 0 \end{bmatrix}$ **l)** $\begin{bmatrix} 2 \\ 4 \\ -1 \\ -3 \end{bmatrix}$

n) $\begin{bmatrix} -2 & -1 & -5 & 1 \\ -4 & -3 & 4 & -1 \\ -4 & 6 & 4 & 2 \\ 4 & 2 & 1 & 2 \end{bmatrix}$ **o)** $\begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$ **p)** $\begin{bmatrix} -2 & 0 & 4 & 0 \\ -2 & 6 & 1 & 0 \\ -4 & 0 & 8 & 0 \\ -4 & 6 & 5 & 0 \end{bmatrix}$

q) $\begin{bmatrix} -9 \\ -11 \end{bmatrix}$ **r)** $\begin{bmatrix} 6 \end{bmatrix}$ **s)** $\begin{bmatrix} 0 & -4 & 3 & 0 \\ 2 & -4 & 0 & 0 \\ 0 & -4 & 3 & 0 \\ 0 & -3 & 0 & 0 \end{bmatrix}$

t) $\begin{bmatrix} 5 & 10 \end{bmatrix}$

7. e 8.

When $\mathbf{A}$ and $\mathbf{B}$ commute.

9.

a) $\mathbf{X} = \mathbf{C}(\mathbf{A}+\mathbf{B})^{-1}$.

b) $\mathbf{X} = (\mathbf{A}+\mathbf{B})^{-1}\mathbf{C}$.

c) $\mathbf{X} = (\mathbf{A}^T)^{-1}$.

d) $\mathbf{X} = \mathbf{A}^{-2}\mathbf{B}^{-1}$.

e) $\mathbf{X} = \mathbf{C}^{-1} - (\mathbf{B}^{-1})^T$.

f) $\mathbf{X} = (\mathbf{A}^{-1} - \mathbf{I})\mathbf{B}$.

g) $\mathbf{X} = \mathbf{A}^{-1} - \mathbf{B}$.

10.
$\left\{ \begin{bmatrix} t-z & -2z \\ z & t \end{bmatrix} \in \mathcal{M}_2 \, : \, z, t \in \mathbb{R} \right\}$.

11.
a) $\mathbf{A} = \begin{bmatrix} -1 & -1/2 \\ 0 & 1/2 \end{bmatrix}$. **b)** $\mathbf{B}^{19} = \begin{bmatrix} -1 & -174763 \\ 0 & 524288 \end{bmatrix}$.

12.
b) $\mathbf{A^{-1}} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.

13.

a)

$$\mathbf{E}(\mathbf{A}) = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \mathbf{E}(\mathbf{B}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \quad \mathbf{E}(\mathbf{C}) = \mathbf{C}$$

$$\mathbf{E}(\mathbf{D}) = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \mathbf{E}(\mathbf{E}) = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \mathbf{E}(\mathbf{F}) = \begin{bmatrix} 1 & 1 & 24 & 1 \\ 0 & 0 & 24 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

b) $r(\mathbf{A}) = 3$. $r(\mathbf{B}) = 2$. $r(\mathbf{C}) = 3$. $r(\mathbf{D}) = 3$. $r(\mathbf{E}) = 3$. $r(\mathbf{F}) = 2$.

14.
for example, $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ e $\mathbf{B} = \begin{bmatrix} 1 & -1 \\ 0 & -1 \end{bmatrix}$.

15.
b) $(\alpha\,\mathbf{A})^{-1} = \dfrac{1}{\alpha}\,\mathbf{A}^{-1}$.

16.
$\mathbf{X} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$.